

Problem-2

Problem 2 - N-Gram Model - 20 Points

During the lecture, we have analyzed the text of Shakespear's Hamlet and built a simple language model, i.e. N-Gram, from it with 1-Markov assumption (the code is `n_gram.py` in the folder Code on Canvas). In this assignment, you have to do the same for the text of Shakespear's Romeo and Juliet. I just copied the text from:

https://shakespeare.mit.edu/romeo_juliet/full.html

into file `Romeo_and_Juliet.txt` (Check folder HW on Canvas). Your tasks are:

1. Clean the data.
2. Construct the vocabulary.
3. Compute the probabilities $p(\text{word})$ and $p(\text{word}|\text{context})$. We assume the 1-Markov assumption.
4. Generate sentences with length $L = 100$ autoregressively with the following first words:
 - Romeo
 - Juliet
 - Love

Important: Remember that you have to implement the proper **sampling** (e.g. you cannot just take the next word with maximum probability). You have to modify the code I gave to you.

5. Create a prompt and ask Large Language Models (LLMs) including Gemini and ChatGPT to create a sentence about Shakespear's Romeo and Juliet that starts with the above words.

Submit your implementation and also your report about your findings.

Ans: In Code and implementation in Assignment-7.ipynb file

Output from Chatgpt

Create a sentence about Shakespeare Romeo and Juliet that starts with the following words.

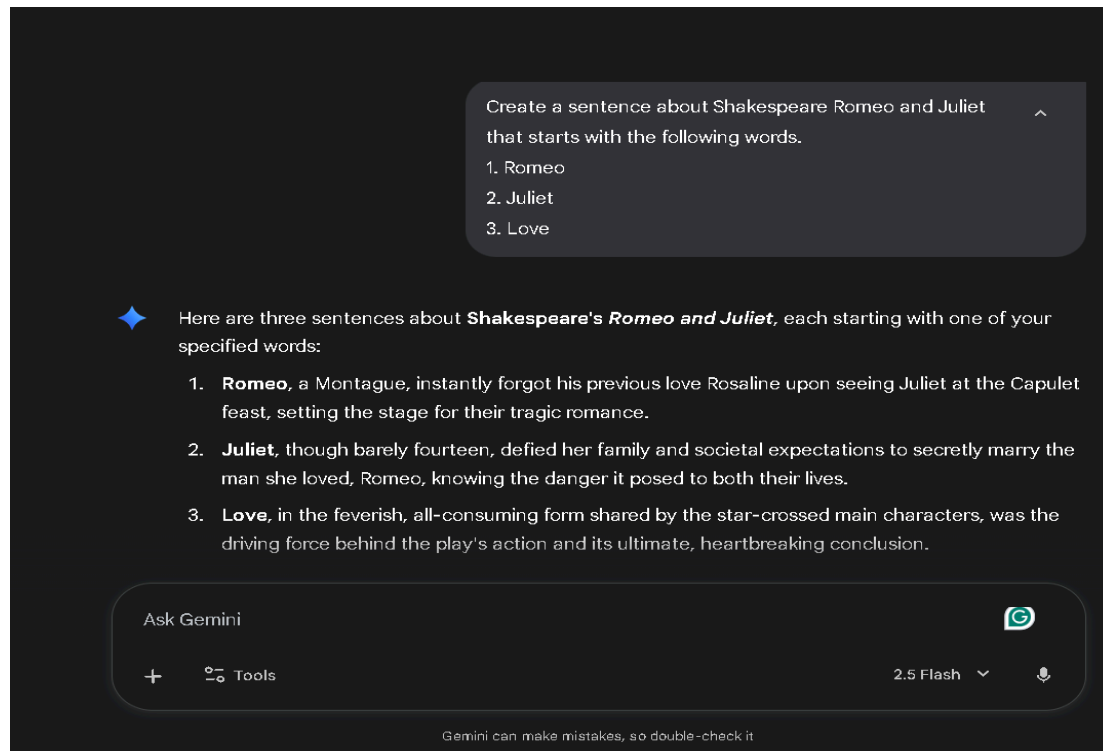
1. Romeo
2. Juliet
3. Love

Sure! Here are sentences for each:

1. Romeo defies his family's feud to pursue his forbidden love for Juliet.
2. Juliet struggles between her loyalty to her family and her devotion to Romeo.
3. Love becomes both the greatest joy and the deepest tragedy in *Romeo and Juliet*.

📄 🌟 💬 ↶ ↷ ...

Output from Gemini



Output that was obtained from N-gram model

**** Printing probabilities ****

probability of word in output file

probability of word|context in output file

Startword = Romeo

Romeo i hither now when i ll quit thy kinsman noble paris every knave i if i care to be fickle fortune by saint an eye and not attend it seems she says nothing what is grown too sore enpierced with romeo stay to a dog a kinsman wherefore storm is in her promotion for thou wilt fall out logs and she had then give me some twenty pieces my man as violently as i am not death and paris beguiled both you will answer i say re enter servingmen capulet alack that on her new mutiny among these many

Startword = Juliet

Juliet wakes and hero hildings and will back to age prince i ll keep off these many orisons to me pronounce it fear too desperate tender kiss by love romeo o what s steel will sure that thy old tear the taste the day never trouble peter o er lawyers fingers goes not so secret nurse romeo i was the villain back thee tidings girl what curious eye or behold twixt them first create o i did not mine friar laurence you that pilgrims ready to be gone so did prompt me nurse behind the hurt cannot sum of love

Startword = Love

Love which late romeo romeo what dares the language servant find delight writ exit juliet o noble earl will lie hath wedded to be merciful open outcry toward our streets hold tybalt s a spirit in this same i know the prince three long spiders legs romeo benvolio with thee beguil d with these gone away romeo meaning for many thousand times the commission of many mine uncle this should have slain whom and may be thoughts of us but the ground knife shall be ready stand in twenty years capulet nurse out of my romeo and love is not

Findings about the output generated by Ngram model and output generated by chatgpt and gemini

1. Outputs generated by Chatgpt and Gemini appear grammatically correct and they make sense but output generated by the N-gram model is not completely grammatically correct. This is because N-gram model just remembers the previous word while generating the next word in the sentence and it ignores the remaining part of the sentence.
2. The N-gram model generated text which includes the main character names anywhere in between the sentences in a random manner which doesnot make sense. On the other hand sentences generated by Chatgpt and Gemini have proper context and meaning in their sentences.
3. N-gram model output doesnot have a flow in the sentences whereas Chatgpt and Gemini used good vocabulary and sentences flow in a sequence.

Problem-3

Problem 3 - Spam Classification - 40 Points

In this assignment, we will build Machine Learning models to classify emails into spam / non-spam (i.e. binary classification) based on a simple embedding. We will work on the Spambase dataset:

<https://archive.ics.uci.edu/dataset/94/spambase>

You can also take a look at my codes that I built spam filtering on this dataset:

<https://github.com/HyTruongSon/Spambase-filtering/>

Your tasks are:

1. Read, clean and organize this dataset into easy-to-read format for Machine Learning (ML) models.
2. Experiment with the following ML models:
 - **Logistic Regression:** https://scikit-learn.org/1.5/modules/generated/sklearn.linear_model.LogisticRegression.html
 - **Decision Tree:** <https://scikit-learn.org/stable/modules/generated/sklearn.tree.DecisionTreeClassifier.html>
 - **Gradient Boosting:** <https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.GradientBoostingClassifier.html>
3. Use **Principal Component Analysis (PCA)** to compress the data (i.e. dimensionality reduction) into $k = 10$ dimensions and report your classification results with Logistic Regression, Decision Tree and Gradient Boosting on the PCA features (i.e. compressed data). You can reuse my code `Lecture_9_PCA_CORA.py` (in the Code folder on Canvas) for PCA.
4. Visualize the embeddings with PCA and t-SNE. You can reuse my code `Lecture_9_PCA_CORA.py` (in the Code folder on Canvas) for these visualizations.

Submit your implementation and also your report about your findings including the accuracy of prediction for each model / case and figures for visualizations.

Output

Features i.e, X: (4601, 57)

Labels i.e y: (4601,)

/usr/local/lib/python3.12/dist-packages/sklearn/linear_model/_logistic.py:465: ConvergenceWarning:

lbfgs failed to converge (status=1):

STOP: TOTAL NO. OF ITERATIONS REACHED LIMIT.

Increase the number of iterations (max_iter) or scale the data as shown in:

<https://scikit-learn.org/stable/modules/preprocessing.html>

Please also refer to the documentation for alternative solver options:

https://scikit-learn.org/stable/modules/linear_model.html#logistic-regression

```
n_iter_i = _check_optimize_result(
```

Without PCA

Accuracy with logistic regression (no PCA): 93.15366224733754

Accuracy with decision tree (no PCA): 99.93479678330797

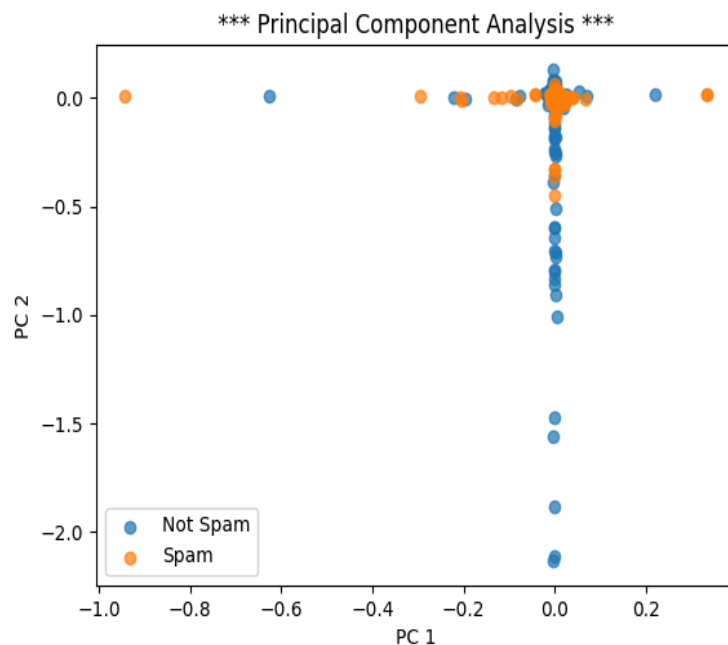
Accuracy with gradient boosting (no PCA): 94.87068028689414

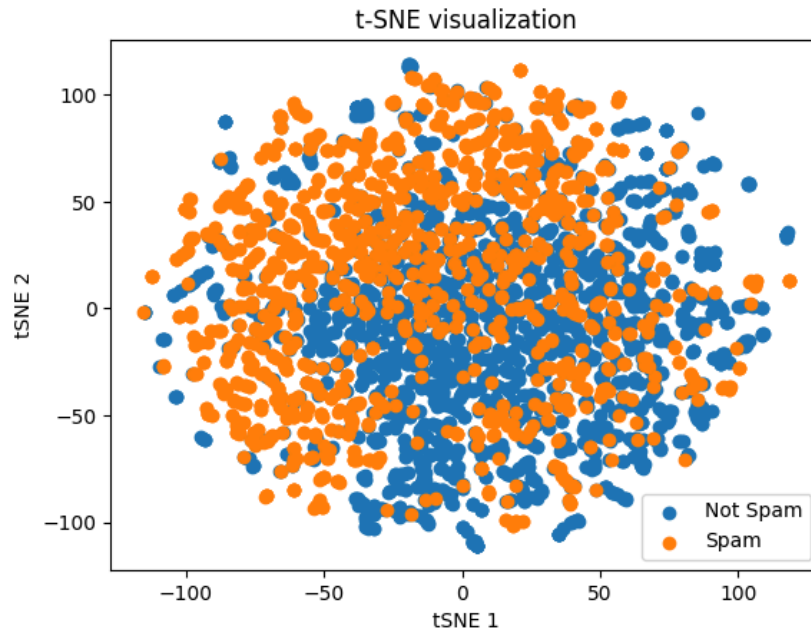
With PCA

Accuracy of Logistic Regression (with PCA compression): 61.812649424038256

Accuracy of Decision Tree (with PCA compression): 99.93479678330797

Accuracy of Gradient boosting (with PCA compression): 83.52532058248207





Done

Problem 4 - LLM API to build a chatbot - 20 Points

Given the Gemini API (`test_gemini.py` in folder Code on Canvas), your task is to build a chatbot. There is **no** need for Graphical User Interface (GUI). The conversation can be on terminal / console. You should continuously ask the LLM to make a summary of conversation and add it to the next prompt of your next question. Submit your implementation and also an example of your conversation / chat with the LLM. Also compare it to the app <https://gemini.google.com/app>. Note: You can use other LLMs for your convenience.

Chat with Gemini

how are you doing today?

◆ I'm doing great, thank you for asking! I'm here and ready to help you with any questions or tasks you have.

How can I assist you today? 🧐

I am wearing a yellow dress . Can you tell me the significance of yellow colour?

◆ That sounds lovely! A yellow dress is a wonderful choice. 🌻

Ask Gemini

+

Tools

2.5 Flash





◆ That sounds lovely! A yellow dress is a wonderful choice. 🌻

The color yellow has a rich and varied **significance** across different cultures and contexts. Here is an overview of its most common meanings: 🔗

🌟 Positive Meanings

Concept	Description
Joy and Happiness	Yellow is often associated with the sun and light, immediately evoking feelings of joy, optimism, and cheerfulness .
Intellect and Learning	It is commonly linked to the mind and the intellect, representing clarity of thought, creativity, and mental alertness .
Energy and Warmth	As a bright and highly visible color, it signifies energy, vibrancy, and stimulation . It is an attention-grabber.
Hope	Like the rising sun, yellow can symbolize a new beginning, hope, and positive anticipation .

 Export to Sheets



Ask Gemini

+

Tools

2.5 Flash





In most personal contexts, wearing a yellow dress typically projects an air of **optimism, energy, and warmth!** 🌟

Is there anything else you'd like to know about the color yellow or your outfit?

how is the weather today?



Ask Gemini



Tools

2.5 Flash



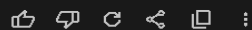
which colour dress am I wearing today?



Based on our previous conversation, you mentioned that you are wearing a **yellow dress** today! 🌟

It was a beautiful choice, and we discussed the cheerful and energetic significance of the color yellow.

Would you like me to suggest an accessory color that complements yellow, or perhaps a different topic?



Ask Gemini



Tools

2.5 Flash



Output from our chatbot

Welcome User! Type 'exit' to quit or else type a question below

Ask me something: How are you?

AIResp: Thanks for asking! As an AI, I don't have feelings or a personal state, but I'm functioning perfectly and am ready to help.

How can I assist you today?

Ask me something: I am wearing a yellow dress? what is the significance of a yellow dress?

AIResp: Of course! That's a wonderful and vibrant choice. A yellow dress is a powerful statement, and its significance is rich and multifaceted, drawing from psychology, culture, and fashion.

Here's a breakdown of what a yellow dress can signify:

1. The Psychology and Emotion of Yellow

At its core, the significance of your yellow dress is tied to the color itself. Yellow is overwhelmingly associated with positive emotions.

- * **Happiness and Joy:** This is the most common association. Yellow is the color of sunshine, sunflowers, and smiley faces. Wearing it can project a sense of joy, optimism, and cheerfulness to others. It can also be a form of "dopamine dressing"—wearing a bright color to intentionally lift your own mood.
- * **Confidence and Attention:** Yellow is a high-visibility color. It naturally draws the eye, so choosing to wear a yellow dress is an inherently confident move. It says that you are not afraid to be seen and that you are comfortable taking center stage.
- * **Energy and Creativity:** As a bright, primary color, yellow is stimulating. It's associated with fresh ideas, intellect, and a burst of energy. It can project a sense of being lively, fun, and creative.

2. Symbolism in Culture and Fashion

Yellow dresses have made their mark in pop culture and carry specific symbolic weight.

- * **Iconic and Memorable:** Think of Belle's ball gown in Disney's **Beauty and the Beast**. Her yellow dress symbolized her inner beauty, intelligence, and grace. Or Kate Hudson's unforgettable yellow silk gown in **How to Lose a Guy in 10 Days**, which represented pure glamour and confidence. Wearing a yellow dress often taps into this iconic, show-stopping energy.
- * **Hope and Remembrance:** The song "Tie a Yellow Ribbon Round the Ole Oak Tree" cemented yellow as a symbol of hope and waiting for a loved one to return. In this sense, it can have a soft, hopeful, and welcoming feel.
- * **Spirituality and Royalty:** In some Eastern cultures, such as Hinduism, yellow is a sacred color associated with divinity and knowledge. Historically, certain shades of yellow dye were expensive to produce, so like purple, it was sometimes associated with wealth and royalty.

3. The Context is Key

The specific significance of *your* yellow dress also depends on a few factors:

* **The Shade of Yellow:**

- * **Bright Lemon/Canary Yellow:** Energetic, bold, and attention-grabbing.
- * **Pastel/Butter Yellow:** Softer, sweeter, and more romantic. It feels gentle and approachable.
- * **Mustard/Ochre Yellow:** More sophisticated, earthy, and subdued. It has an artistic, intellectual feel.
- * **Gold/Marigold:** Luxurious, celebratory, and warm.

* **The Style of the Dress:**

- * **A Flowy Sundress:** Suggests a carefree, happy, and relaxed attitude—perfect for a summer day.
- * **A Formal Gown:** Bold, glamorous, and unforgettable. It's a choice for a major event where you want to shine.
- * **A Structured Sheath Dress:** A powerful and confident choice for a work or professional event. It says you're creative and not afraid to stand out.

In Summary: What Does Your Dress Say?

When you wear a yellow dress, you are most likely projecting:

- * **Positivity and warmth.**
- * **Confidence and self-assurance.**
- * **An approachable and joyful personality.**
- * **A vibrant, energetic spirit.**

Ultimately, the most important significance is how it makes *you* feel. If you feel happy, confident, and bright in your yellow dress, that's exactly the energy you are radiating to the world. It's a fantastic choice

printing chat summary:

Summarised: Of course. Here is a summary of the conversation:

The conversation began with the user asking the AI how it was. The AI responded that as an AI, it doesn't have feelings but is functioning perfectly and ready to help.

The user then asked about the significance of wearing a yellow dress. The AI provided a detailed explanation, covering four main points:

1. **Psychology:** A yellow dress is associated with positive emotions like happiness, joy, and optimism. It's an attention-grabbing color that projects confidence and energy.
2. **Culture:** It has iconic cultural significance, often symbolizing glamour and beauty (like in movies) or hope and remembrance.
3. **Context:** The specific meaning depends on the shade of yellow (e.g., bright vs. pastel) and the style of the dress (e.g., casual vs. formal).
4. **Overall Meaning:** In summary, wearing a yellow dress projects positivity and self-assurance, but the most important significance is how it makes the person wearing it feel.

Ask me something: how is the weather today

AIResp: I can't give you the current weather for your specific area because I don't know your location.

However, if you tell me your city (and state or country), I can look up the forecast for you right now

printing chat summary:

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The conversation began with the user asking about the significance of wearing a yellow dress. The AI provided a detailed explanation, covering that a yellow dress can signify:

- * **Positive emotions** like happiness, joy, and confidence.
- * **Cultural and fashion symbolism**, referencing iconic movie dresses and hope.
- * How the meaning can change based on the specific **shade of yellow and style of the dress**.

The topic then switched, and the user asked for the weather. The AI responded that it could not provide the weather without knowing the user's location and asked for their city to give a forecast.

Ask me something: which colour dress am I wearing today?

AIResp: That's a fun question!

As an AI, I don't have eyes or cameras, so I can't see you to know what color dress you're wearing.

However, based on our previous conversation, I'll make a guess: is it a yellow one? You'll have to tell me if I'm right

printing chat summary:

Summarised: Based on the conversation provided, here is a summary:

The user tests the AI's awareness of the user's real-world, personal environment.

1. **First Question (Weather):** The user asks for the weather. The AI explains its limitation—it doesn't know the user's location—and then offers a solution by asking the user to provide their city.
2. **Second Question (Dress Color):** The user asks a question that requires physical sight. The AI again explains its limitation—it has no eyes—but responds in a more playful and engaging manner by making a guess.

In short, the conversation highlights the AI's inability to access personal, real-time physical information about the user and how it communicates these limitations.

Ask me something: exit

Goodbye! Have a great day.

Previous Messages :

UserInput: How are you?

AIResponseObtained: Thanks for asking! As an AI, I don't have feelings or a personal state, but I'm functioning perfectly and am ready to help.

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Summarised Content:

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In short, the conversation highlights the AI's inability to access personal, real-time physical information about the user and how it communicates these limitations.

Observations and Comparison of LLM v/s Gemini

1. As highlighted in blue when asked about the dress later it correctly predicted that it was yellow. But we have a limit of summary of previous 4 conversations here. But for Gemini it has inbuilt mechanism of remembering the previous context while answering a question.
2. The LLM implementation always gave big big responses for every question asked but Gemini's responses were concise and to the point.
3. When asked about the weather, Gemini responded with a visual representation of weather and the temperature value but our LLM just prints a long piece of text.

Problem 5 - LLM API to get embeddings - 10 Points

Given the Gemini API (test_gemini.py in folder Code on Canvas), your task is to extract the sentence embedding for the 4 sentences in Problem 1 of this homework. You can limit the embedding size to 20. Submit your implementation and your report about the results. **Note:** You can use other LLMs for your convenience.

Output:

Sentence : I study Computer Science

[-0.002668373, -0.007002924, -0.09527657, -0.0036271424, 0.024797544, 0.019541493, 0.002165475, 0.028409896, -0.018734418, 0.016721116, 0.0062950053, 0.010313386, 0.06567957, -0.014763642, -0.004515892, -0.019287439, 0.025833864, 0.036357276, -0.09585416, 0.035041355]

Sentence : I learn Computer Engineering

[0.0013030831, -0.011818702, -0.07862751, 0.014445912, 0.029969627, 0.0011539815, 0.0054108994, 0.009536284, -0.017068464, 0.020247178, 0.0008837594, -0.0075590354, 0.064006664, -0.010558932, -0.0040601776, -0.038867332, 0.035677508, 0.03340209, -0.1041233, 0.024407284]

Sentence : We learn Computer Science

[-0.0018595563, 0.014690126, -0.062269825, -0.026924247, 0.018988924, 0.007066536, -0.012727012, 0.0010292843, -0.0029372117, 0.030724183, 0.009297237, -0.0032500823, 0.08967312, 0.013003159, -0.0017132163, -0.014582471, 0.04929346, 0.018904757, -0.11978554, 0.03394048]

Sentence : We study Computer Science and Computer Engineering

[-0.014612368, 0.023956291, -0.06386977, -0.022939194, 0.023786763, -0.0082291225, 0.032828532, -0.007900169, 0.00092211086, 0.021985745, 0.007519039, 0.007858009, 0.07613782, 0.01316484, 0.010115148, 0.008172, 0.050208066, 0.024397511, -0.12598003, 0.022480212]